The economic impact of climate change on coral reef in the Main Hawaiian Islands

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June 1, 2024

Abstract

Coral reefs, characterized by their rich diversity, are productive ecosystems contributing to the provision of a wide range of ecosystem services, including recreation, coastal protection, and marine biodiversity. Climate change impacts, including ocean warming and acidification, pose a significant threat to coral reefs and the associated provisioning of ecosystem services. The variability of these impacts underlines the need to develop more spatially explicit tools in coastal ecosystem management that integrate and assess potential ecological and socio-economic outcomes. To address this, a spatially explicit predictive ecological model is applied to project changes in coral reef cover, using downscaled data from Shared Socioeconomic Pathway (SSP) climate scenarios. Based on these projections, welfare impacts of changes in recreational value are estimated across different populations and landscapes. Cumulative welfare losses for Hawaii residents range from $2.1 to $3.3 billion by 2100. Welfare loses are occasionally higher under lower SSP scenarios due to higher predicted populations and more individuals experience the loss at the state level. The approach incorporates site-specific characteristics, income distribution, and projected regional population growth to connect ecological change with welfare outcomes. EJScreen is used to assess variation in welfare impacts across communities, identifying disadvantaged populations based on demographic and environmental indicators such as poverty, minority status, and exposure to environmental risks. These findings can inform policy and resource allocation by supporting ecosystem management strategies that account for both ecological dynamics and community-level socio-economic conditions.

Keywords: Coral reefs; Climate change; Integrated Ecosystem Modeling; Economic Valuation

JEL codes: Q26; Q51; Q54; Q57

Highlights

* Disadvantage communities estimated to consistently have higher per person welfare losses than other communities.
* Welfare losses for Hawaii residential range from $2.1 to $3.3 billion by 2100.
* Welfare losses are higher in certain SSP1-2 scenarios due to projected population growth, which results in more individuals experiencing welfare loss.

This work was supported by the National Oceanic and Atmospheric Administration (NOAA) Ocean Acidificaiton Project (OAP).

Introduction

Coral reefs cover just 0.2% of the seafloor, yet support 25% of marine species (Souter et al., 2021). These biodiverse ecosystems provide a wide array of ecosystem services, including coastal protection, food, and economic security for a billion people (Cesar & Beukering, 2004; Colt & Knapp, 2016; Reguero et al., 2021). Acting as natural barriers, they protect tropical coasts and islands from waves (Ferrario et al., 2014; Harris et al., 2018; Pascal et al., 2016; Reguero et al., 2021). Coral reef fisheries contribute significantly to the global economy, supporting millions of livelihoods (Fernandes et al., 2017; Speers et al., 2016). Additionally, the recreation and tourism sectors heavily depend on the quality of coral reefs, with 30% of the world's reefs holding value for tourism (Spalding et al., 2017).

Coral reefs face increasing challenges in the face of anthropogenic climate change due to interconnected factors such as ocean acidification and rising sea temperatures (Hoegh-Guldberg et al., 2017; Pandolfi et al., 2011). Ocean acidification impedes coral growth and structural integrity by hindering their ability to build skeletons, while rising sea temperatures contribute to increased coral bleaching events (Kleypas & Yates, 2009). The Red List Index, which tracks the trends of specific taxonomic groups, notes the decline of corals as particularly precipitous (IUCN, 2018). Live coral cover on reefs has nearly halved in the past 150 years and is projected to virtually disappear within the century even with optimistic climate mitigation (Díaz et al., 2019; Frieler et al., 2013). Land-based stressors, such as pollution, and overfishing diminish the ability of reefs to survive warming and acidifying oceans (Andrello et al., 2022; Good & Bahr, 2021; McLean et al., 2016).

These climate-induced changes pose risks to the economic values communities derive from healthy coral reefs, particularly impacting provisioning services such as fisheries (Armstrong et al., 2012; Estrada & Botzen, 2021; Moore & Fuller, 2022; Narita et al., 2012), regulating services like coastal protection (Harris et al., 2018; Pascal et al., 2016), and cultural services including tourism and recreational opportunities (Armstrong et al., 2012; Cesar & Beukering, 2004; Fezzi et al., 2023; Peng & Oleson, 2017; Spalding et al., 2017, 2017). Despite the dire outlook for coral reefs and the impending loss of the valuable services they provide to society, these fragile and important ecosystems are rarely managed optimally because their benefits are not captured in traditional markets (Mackie & Abraham, 2005; NSTC, 2023; Tallis et al., 2024). Given the significant impacts of climate change and human dependence on coral reefs, understanding their

vulnerability and assessing the economic consequences of their degradation is critical for effective management decisions. This approach enables environmental managers to develop strategies that better ensure the long-term resilience of coral reefs, safeguarding the well-being of both the ecosystems and the communities that depend on them. Importantly, climate change impacts on coral reefs will vary spatially, causing different regions and communities to experience varying degrees of degradation or resilience (Frieler et al., 2013; Souter et al., 2021). For environmental managers, this calls for a planning approach that integrates both the ecological and economic consequences of spatial variability, thereby supporting the development of targeted management strategies (Bateman et al., 2013; Ferrini et al., 2015).

Research is key to guiding planning that looks at both the ecological health of coral reefs and their importance to people. Previous studies have coupled ecological models to economic valuation to calculate the economic implications of changing reefs (Fezzi et al., 2023; Lane et al., 2015; Oleson et al., 2020; Peng & Oleson, 2017) and the benefits of management actions (Hafezi et al., 2020, 2021; Pathak et al., 2021; Reguero et al., 2021; Suggett et al., 2023). These studies are often regional in scale because the available data and climate scenarios usually lack the detail needed to inform local reef management (McClanahan, 2024; van Hooidonk et al., 2016). Effective coral reef management, especially across biogeographically and socio-culturally diverse islands, depends on a localized, place-based approach (Jennings et al., 2024; Wedding et al., 2024), but only recently have downscaled projections of climate change become available for the MHI at scales relevant to local management of reefs (Asner et al., 2020).

This study addresses a research gap by integrating a spatially explicit ecological model (Audzijonyte et al., 2019) that projects long-term climate-driven changes in coral reefs with economic valuations of recreational service changes. We account for site-specific variability and assess welfare trade-offs under various Shared Socioeconomic Pathways (SSPs) (O'Neill et al., 2014). More specifically, the approach begins by defining recreational beach locations, available amenities, and ecological conditions at each site across all main Hawaiian Islands (MHI), including Maui, Kaua‘i, O‘ahu, Lana‘i, Moloka‘i, and Hawai‘i Island. A benefit-function transfer exercise, based on the multi-site travel cost model estimated for Maui by Fezzi et al. (2023) is then applied to estimate recreational values for residents of the MHI at each site. These values are used to assess welfare impacts resulting from projected coral reef losses under different SSP scenarios at site-specific locations using the Atlantis model. Furthermore, welfare loss heterogeneity is assessed by

overlaying indicators that identify historically disadvantaged communities, emphasizing critical environmental justice concerns. This approach demonstrates how climate-driven welfare impacts disproportionately burden these vulnerable populations. The analysis employs EJScreen, a recently developed tool that integrates environmental and demographic data—including poverty rates, minority status, and exposure to environmental hazards—to better understand the distributional implications of localized welfare changes.

These findings offer spatially relevant insights for decision-makers to maximize social welfare by guiding targeted restoration and protection by identifying vulnerable communities under future ecological changes. We find welfare losses across disadvantage communities are often larger than those in non-disadvantage communities. This finding supports ongoing efforts by government agencies to identify heterogenous impacts across vulnerable communities. Furthermore, ecological modeling suggests coral reefs face a stark future even under the most optimistic climate emission scenarios (shared socioeconomic pathways 1-2). Coupled the most optimistic climate scenario and its corresponding regional population growth, welfare changes are higher under optimistic scenarios because there are more people experiencing welfare losses due to the loss of coral reefs within the nearshore environment.

Data

Recreational Sites

Beach recreational sites across the main Hawaiian Islands (Maui, Oʻahu, Kauaʻi, Molokaʻi, Lanaʻi, and the island of Hawaiʻi, or "Big Island") were identified by manual merging existing spatial datasets (e.g., Hawaiʻi Statewide GIS program[1]) coastal sites listed online with Google Places. All potential coastal recreational sites were considered. Each site was manually cleaned by cross-checking with tourism guides and satellite imagery to verify site existence and locations across the islands.[2] Figure 1 illustrates the identified coastal sites and residential areas, as well as areas under various forms of management by the Division of Aquatic Resources (DAR). These areas are administered by DAR with a goal to manage, conserve, and restore the state's unique aquatic resources and ecosystems for present and future generations, and thus important to monitor for potential threats under different climate scenarios.

[1] https://geoportal.hawaii.gov/
[2] https://www.gohawaii.com/; https://www.hawaiibeaches.com/

Recreational site characteristics (e.g., amenities, biophysical attributes, beach length) were collected to support a benefit function transfer. Activities and amenities considered included surfing, swimming, snorkeling, presence of lifeguards, showers, parking, restrooms, and sandy or rocky coastlines, identified using tourist guides and satellite imagery. Additionally, each coastal site was characterized by biophysical attributes such as coastline length, average coral cover, and average fish biomass. For site specific biophysical attributes, a 300-meter buffer zone to calculate the average coral cover and resource fish biomass representing the geographic data. The precent of coral cover of each site was determined utilizing (Asner et al., 2020) and the average fish biomass data for each location was sourced from the Marine Biogeographic Assessment of the Main Hawaiian Islands (Costa & Kendall, 2016). Table 1 summarizes the characteristics of the coastal sites by island.

Demographics & Travel Cost

The 2020 Census block-level population estimates for each island, collected by the Census Bureau (Figure 1), were used.[3] Income data, matched at the census tract level to each subgroup population grid, was calibrated to the median income of the original population. Climate-simulated scenarios are combined with the environmental justice screening tool to better identify disadvantaged communities and inform more refined policies. This tool provides data on the various burdens faced by communities by merging social, economic and environmental data and screening for exposure risk. Social and economic data is derived from U.S. Census Bureau American Community Survey while environmental data include air pollution, climate risk, wastewater discharge, health outcomes, etc. (EPA, 2024).

To build the scenarios, projected increases in human population growth over the 21st century are accounted for using population projections for the state of Hawai‘i consistent with the Shared Socioeconomic Pathways (Jiang et al., 2020; Zoraghein & O’Neill, 2020). The US is expected to experience a medium level of population growth in both SSP1 ('Sustainability') and SSP2 ('Middle of the Road') which is driven by medium levels of fertility, mortality, and international migration. Population growth is low in SSP3 ('Regional Rivalry') due to low fertility and international migration along with high mortality.

[3] http://census.gov/geographies/reference-maps/2020/geo/2020-census-block-maps.html

For each island, a gridded population density map was created at a 1-kilometer squared (1 km^2) scale. This gridding process allows for the calculation of distance and travel time from each grid to each site located on the same island. This travel cost method approach increases the precision of the travel decision for micro-population within each island (Table 1, Distance per Island). Distance and travel times to each site from each grid are calculated using the Open Source Routing Machine (OSRM).(Giraud, 2022). OSRM's advanced routing algorithms ensure a cost-effective, accurate, and efficient solution for estimating travel times comparable to more costly options such as Google Distance Matrix API and ArcGIS Network Analysis (Fu et al., 2023). In line with Fezzi et al. (2023), round-trip travel costs are calculated by combining monetary expenses (assuming a fuel efficiency of 20 miles per gallon and a fuel price of $4.30 per gallon) with the value of travel time savings, estimated at $18.10 per hour. This time value corresponds to three-quarters of the average wage rate, following the approach of (Fezzi et al., 2014), based on an annual workload of 2,000 hours (Haab & McConnell, 2002; Leggett et al., 2018).

Methods

Economic Modeling

To estimate welfare losses resulting from changes in ecological conditions under different climate scenarios, a benefit transfer approach is applied using a value function estimated for Maui residents by Fezzi et al. (2023). Benefit transfer involves transferring valuation estimates from a study site to one or more policy sites (Johnston & Wainger, 2015). The function transfer uses parameters derived from a multi-site travel cost (TC) model. As a detailed description of the model and the data used for estimation is provided by Fezzi et al. (2023), a brief synopsis is provided here. An online survey was sent out to more than 300 adult residents of Maui Island in November and December 2017 to obtain information about their daytrips to outdoor recreation sites among. The sample was obtained using Qualtrics[4], and respondents were incentivized to participate through gift cards and coupons. The residents took nearly 3,000 trips to more than 170 outdoor sites located across the island of Maui.

Fezzi et al. (2023) estimate preference parameters for amenities and environmental attributes at coastal sites in Maui using a multi-site TC model. The multi-site TC model is a workhorse for modeling recreational choices and is often applied to the valuation of coastal

[4] www.qualtrics.com

recreation (Dundas & Von Haefen, 2020; Leggett et al., 2018; Lopes & Whitehead, 2023; Phaneuf & Smith, 2005; Whitehead et al., 2010). The random utility model (RUM) forms the foundation of multi-site TC models, assuming that individuals maximize utility when choosing which recreation site to visit based on factors such as travel cost, site amenities, and environmental attributes. In the model, each day (choice occasion, *t*) and each individual n will maximize their utility by choosing which site to visit between I different sites (which also include the option of not partaking in outdoor recreation, which is referred to as "remain at home"). From this maximization problem, the indirect utility function associated to choosing site i can be written as the sum of a deterministic component (V_{nit}) and a stochastic component (ϵ_{nit}). The deterministic component defines observable components influencing utility, whereas the stochastic component capture factors explaining utility that are unobserved by the researcher but observable to the individual. The indirect utility function can be written as:

$$U_{nit} = V_{nit} + \epsilon_{nit} = \theta tc_{ni} + \beta' x_i + \epsilon_{nit}, \tag{1}$$

where tc_{ni} is the travel cost from the individual's home to site i (which also includes the value of travel time, which is set to ¾ of the average wage rate following (Fezzi et al., 2014)), and is zero if the individual chooses to stay at home, x_i is a vector of site characteristics (e.g., coral cover, fish biodiversity, presence of a amenities). Lastly, θ and β are preferences parameters to be estimated for marginal changes in TC and site characteristics, respectively. Assuming that the stochastic components are independent and identically distributed as extreme value type I (Gumbel), the conditional logit model is obtained, expressing the probability of individual n choosing to visit site i in choice occasion t as:

$$\text{Prob}(y_{nit}) = \prod_{t=1}^{T} \frac{\exp(\theta tc_{ni} + \beta' x_i)}{\sum_{j=0}^{J} \exp(\theta tc_{nj} + \beta' x_j)}., \tag{2}$$

Following the standard economic assumption of linear utility and additive income (Hanemann, 1999), the compensating variation in welfare for individual n, associated with changes in site quality for each recreational choice occasion, is given by:

$$\theta^{-1} \ln\left(\frac{\sum_{i=0}^{I} \exp(V_{ni}^{*})}{\sum_{i=0}^{I} \exp(V_{ni})}\right), \tag{3}$$

where V_{ni}^{*} defines the attributes of the sites after the quality change and can be used to calculate the welfare changes for each individual in a sample. For purposes of this study, the aggregated welfare is calculated by the number of people resident in each 1km squared grid. Such welfare changes represent only use values derived from recreation, and can be interpreted as the Willingness to Pay (WTP) of each individual for avoiding the changes in quality from V_{ni} to V_{ni}^{*} across all possible sites I.

The model by Fezzi et al. (2023) estimates welfare changes of quality changes for the sites on the island of Maui. To extend the results to the other islands, a value function transfer approach is used to calculate welfare changes for sites on those islands. The value function transfer is then defined as:

$$\overline{WTP}_{ir}^{BT} = f\left(\left[x_{ir}, x_{js}\right], \hat{\beta}_{js}\right) \tag{4}$$

In this equation, $\overline{WTP}_{ir}^{BT}$ represents the adjusted predicted welfare estimate for population group r at site i, based on transferring values from the original population sample s, who visited sites j on the island of Maui. The variables x_{ir} and x_{js} describe the characteristics of the population r and s, as well as characteristics of sites i and j, respectively. The term $\hat{\beta}_{js}$ refers to the vector of estimated preference parameters from the original study, based on a vector of sites j, which are applied to the new sites i. Each subgroup population is defined in km gridded squares for fine scale resolution. Results are further stratified by income. For each income level, the following formula is used:

$$\widehat{WTP}_{ir}^{BT} = \overline{WTP}_{ir}^{BT} \cdot \left(\frac{y_r}{y_s}\right)^{\eta}, \tag{5}$$

where y_r is median household income for subgroup population r matched to each census tract, y_s is the original reported median household income from the study site, and η is the income elasticity, which is assumed to be 1 (Drupp et al., 2024). This exercise applies transferable value functions that are economically robust and account for site-specific characteristics in a spatially explicit manner (Bateman et al., 2011).

Leave-one-out Cross Validation

To examine the model accuracy of the benefit transfer function, we perform a cross-validation known as leave-one-out cross validation (LOOCV) (Hansen, 2022). LOOCV is a widely

used technique to assess a statistical model prediction when new data are not available and is commonly applied to estimate model performance and to compare among different models (Hansen, 2022). The process involves dividing the available data into multiple sets to train a model on portion of the data and then testing it on the excluded set and then repeating this process iteratively. The original model was analyzed and outcomes were predicted using leave-one-out validation cross-check (LOOVC) by iteratively excluding individual sites. This process produced an average root mean squared error (RMSE) of 0.18. Given that the original model employed an aggregated logit approach, this RMSE indicates reasonably accurate predictions.

Coral Reef Modeling

The Atlantis ecosystem model framework (Fulton et al., 2004) , previously tailored for the MHI (Weijerman, 2020) , was used to simulate nearshore ecosystem outcomes through the end of the century. Atlantis is a spatially explicit, three-dimensional, deterministic model that accounts for biophysical dynamics, predator-prey relationships, and species life histories. The model's spatial structure captures differences in marine habitats, biological communities, and management by dividing the area into user-defined polygons. For the Main Hawaiian Islands, 92 polygons were designed based on regions with similar biogeographic and oceanographic conditions and correspond to fishery reporting zones. These polygons are shown in Figure 2.

The Atlantis was then forced through incorporating global climate data from CMIP6 to simulate three different climate scenarios, Socioeconomic Pathways scenarios SSPs 1-3. The scenarios representing low, moderate, and high emission pathways and then simulate potential outcomes for the Hawai'i region. The Atlantis modeling predicts changes in coral biomass within each polygon and coral types modeled include branching and massive *Pocillopora, Porites branching, Porites massive, Montipora, Leptoseris*. To integrate ecological conditions into the economic model, the percentage change in coral biomass within each Atlantis polygon is calculated

$$C_{pt} = \frac{\sum V_{pt}}{\sum V_{pt=0}} \tag{6}$$

Where V represent the total biomass of coral within Atlantis in polygon, p, at time, t, all over the coral biomass baseline for the baseline period of 2015-2019 which represents the economic survey timeframe, $t = 0$. The result is C which represents the percentages change in coral within is

polygon in polygon, p, at time, t. The projected changes are aggregated into yearly timesteps which are projected annually up to 2100.

Integration

Figure 2 illustrates the integration of the modeling projections. The Atlantis biophysical model predicts ecological changes in the marine environment around the MHI based on human, ecological, hydrographic and climate forcing. The changes projected by Atlantis are translated into percentage changes in the coral reef within each polygon (Eq. 6). The change overtime is applied to each coastal site corresponding to the polygon in which the site resides. Each site has downscaled coral cover estimated from Lidar data in the nearshore environment (Asner et al., 2020; Fezzi et al., 2023) and amenities following (Fezzi et al., 2023).

The economic welfare measure is applied across population within 1km2 grids across each island. Economic impacts over time were quantified by calculating the compensating variation associated with change in ecological condition at each site. The welfare calculation is estimated for each year time step and can represented across spatial landscape or aggregated across islands. The aggregated values are coupled with yearly population growth estimated using state level population estimates under the corresponding climate scenarios.

Results

Welfare impacts and projected changes in nearshore coral reef conditions are integrated at site-specific locations over time by converting Eq. (2) into an annual welfare measure, calculated by summing the value of a single daily choice occasion across the year and aggregating by the grid-level population. Welfare measures are adjusted by median income at the corresponding tract level. Figures 3 map the welfare under SSP1 & 3 in the near (Figure 3.a 2030), mid (Figure 3.b 2050), and late century (Figure 3.c 2100). Interpretation of these estimates should be thought of as how much the current resident would need to be compensated for the welfare losses cause by the decline in recreational ecosystem services due to the degradation of coastal ecosystem quality. Each estimate demonstrates potential vulnerable communities under future ecological changes where annual welfare losses become concentrated at locations with recreational locations and loses of coral cover are projected.

The nearshore environment across the Hawaiian Islands is heterogeneous, with climate scenario projections under SSP1 and SSP3 revealing distinct spatial patterns of ecological and

economic change. Both scenarios indicate that ecological changes to coral reefs are expected to be severe, with substantial declines projected by mid-century and near collapse by the late century. These findings align with previous research on coral reef degradation under climate change scenarios in the Hawaiian Islands (Lane et al., 2015). Although both climate scenarios express large welfare losses, the analysis does highlight a few potential differences across the scenarios. A key distinction between these scenarios is that SSP3 consistently produces larger, more isolated areas of welfare loss and coral cover decline. These locations a can be found in 2030 on the southwest of the island of Hawaiʻi, around the island of Oʻahu in 2050, and various locations in the 2100.

Potentially vulnerable locations are further highlighted by identifying disadvantaged communities using the EJScreen Toolkit (EPA, 2024; Mullen et al., 2023). This tool provides detailed information at a high geographic resolution for both demographic and environmental indicators to identify areas where environmental burdens disproportionately affect low-income and minority populations. Using the Table 2 presents the annual welfare losses per person by island and community, revealing significant disparities. On Maui, disadvantaged communities near Wailuku and Kahului bear consistently higher welfare losses compared to other areas on the island. Similarly, the entire islands of Lānaʻi and Molokaʻi, classified as disadvantaged, face heightened concerns regarding heterogenous impacts across the region. On Oʻahu and Kauaʻi, welfare losses start in disadvantage communities and overtime are outpaced by other communities. In contrast, communities on Hawaiʻi Island not identified as disadvantaged exhibit the most significant differences, with ecological projections indicating substantial nearshore environmental losses on the island's east side (Figures 3) and moderate impacts near identified communities until 2100.

Lastly, a localized fit of annual welfare losses was estimated by island. Projecting these values into the future requires determining an appropriate discount rate. This process translates future society benefits into their present value and inevitably diminishes the long-term environmental damages associated with the welfare future generation. The decision of discount rate can have implications for the value of future flows into long-term decision-making of climate change (Nordhaus, 2010, 2017; Stern, 2007). The empirical setting involves distant future generation with uncertainty as well as involving ecosystem services of a Red-List-Index species (Baumgärtner et al., 2015; Newell & Pizer, 2003). Costanza et al., (2021) propose a pluristic discounting for long-term discounting for ecosystem services which rely on relative contribution

from capital. The setting includes recreational services of an ecosystem service which rely on a mix of contribution from human, social, natural and built capital and thus a fixed discount rate is likely not be appropriate for long term appraisal and incorporation into decision-making (Costanza et al., 2021). A pluralistic discounting approach, as presented in Costanza et al., (2021) is applied using the recreational value composition: 40% human, 10% social, 30% natural, and 20% built capital, with corresponding discount rates of 0%, 0%, 3%, and 10%, respectively, for each asset. A declining discount rate is also considered for the long-term time horizon, following the social rate of time preference proposed in OMB Circular A-4 (Office of Management & Budget, 2023). Lastly, population projects for Hawai'i based on each climate scenario interpolated to a yearly scale and matched to welfare estimates (O'Neill et al., 2014). Population estimates to 2100 in Hawaii are projected to increase across SSP1 and SSP2 and stagnate under SSP3 due to assumptions underlying developed economies migration and fertility in the United States.

Figure 4 shows annual welfare loses across island. The highest welfare loses across are scenarios are estimated use pluralistic discounting accounting for capital assets contribution for recreational ecosystem services. The lower welfare loses uses the long term discounting suggest by Circular A-4 (Office of Management & Budget, 2023). SSP1 and SSP2 suggest potential higher welfare losses driven by the higher population growth to Hawaii under the scenarios. Intuitively, under these scenarios there are more individuals to experience the projected loss of coral reefs by the end of the century.

The ecological modeling in Atlantis does suggests areas of recovery of coral reefs in the nearshore environment on islands like Kaua'i, Mau'i and O'ahu. Those times of recovery are the driving force of positive welfare improvements in the early years of welfare. Those moments of recovery, however, are short lived and are followed by a steep welfare decline across islands due to declining percentage of coral reefs in the area. While the timing of the decline varies across island, all islands and model suggest stark declines by the mid-century. The other noticeable trend in the projections is the recovery trend in the lower bound estimate in welfare estimates in the end of the century and is driven by the discount without consideration of capital assets contribution of the ecosystem service. Lastly, the cumulative welfare loses across all islands range from from $2.1 to $2.7 billion (2024 dollars) by 2100 using OMB Circular A-4 discounting and $2.5 to $3.3 billion (2024 dollars) by 2100 using a pluralistic discounting accounting for capital type.

Discussion

Coral reefs face significant threats from climate change, through drivers such as ocean acidification and bleaching from higher sea temperatures, resulting in reduced coral cover (Hoegh-Guldberg et al., 2017; Munday et al., 2008). Reduced coral cover can bring significant economic consequences (Chen et al., 2015; Speers et al., 2016), including the loss of recreational values (Brander et al., 2007, 2012). These values are not entirely reflected in market transactions, and are thus subject to the use of nonmarket valuation methods to assess their economic importance, otherwise, they risk being undervalued or ignored in policy formulation. Therefore, assessing future recreational value losses due to coral cover decline from climate change requires combining ecological models with nonmarket valuation ones. This integrated approach helps inform policy by providing insights into both the ecological consequences and the social welfare impacts of coral reef degradation. However, integrated models have inherent uncertainties (Baustert et al., 2018). These uncertainties, along with the main findings, are discussed in the following section.

The main contribution to the literature arises from spatially heterogenous estimation of the economic impacts for Hawai'i residents. The analysis estimated that the welfare losses for all residential from changes in coral cover are between $2.1 to $3.3 billion for the year 2100 in 2024 dollars. These estimates are lower than those projected in previous studies for Hawai'i. Lane et al. (2015) which suggest a loss of $15.2 to $23.1 billion in 2024 dollars by the year 2100. The differences in estimates are driven by the scope, the benefit transfer method and the discounting rate. The analysis restricted its scope to residents of Hawaii by transferring a functional model based off amenities and ecological conditions of recreational site by the year 2100. Lane et al. (2015) uses a value transfer and the annual estimates of snorkeling and diving trips for the market of both residents and tourist by using an aggregated trip reported from Cesar & Beukering, (2004) and a fixed discount rate of 3%. The number of residents in Lane et al. (2015) accounted for around 10% of all estimated trips (Cesar & Beukering, 2004; Lane et al., 2015). Thus, the method suggests values associated with the local population are 38% to 43% higher than the welfare estimates compared to Lane et al. (2015). Inclusion of the welfare impacts to tourist would give us a more comparable estimate, however, researcher into the spatial welfare measures for tourist estimates is left for further research.

The analysis assumes linear effects, with the estimated future reductions under each climate scenario interpreted as marginal changes (Dugstad et al., 2021; Johnston & Wainger, 2015).

However, the true functional form between welfare and coral reefs degradation has uncertain inherent nonlinear relationships. Coral reef are predicted to experience extreme high biodiversity losses under even optimistic climate scenarios (Fabricius et al., 2011; Frieler et al., 2013; Hoegh-Guldberg et al., 2007). The willingness to pay for the ecosystem services corals provide will increase even more as coral reefs become scarcer and thus the estimates in the long term are likely highly undervaluing the contributions of the function ecosystem and its contribution to society (Drupp et al., 2024). This attempts to address this concern by employing both a declining and pluralistic discount rate in the annual welfare estimates. This process produces large bounded areas for each future scenario (Figure 4). Declining discount rates apply a single rate to the entire stream of costs and benefits over time, implicitly assuming uniform intertemporal preferences for all aspects of recreational services, regardless of their source. However, this assumption overlooks important distinctions in the types of capital contributing to recreational value. As this analysis focuses on recreation associated with a specific form of natural capital, coral reefs, a pluralistic discounting approach is also considered. This method allows for differentiation among the characteristics and behaviors of various capital types embedded in the model. Because recreational value is closely tied to the experience of a healthy natural environment, accounting for capital type better reflects the true value of foregone future benefits.

This study explores a benefit function transfer approach to estimate values for other sites across the main Hawaiian Islands. Importantly, this transfer approach is generally preferred over other benefit transfer techniques, such as unit value transfer, as it allows for accounting for differences in site characteristics (Rosenberger & Loomis, 2017). The leave-one-out cross-validation average root mean squared error (RMSE) indicates good predictive accuracy. However, the approach assumes that preferences for recreational services are constant across island populations while adjusting for income differences. Preferences for recreational services may vary across different island populations. However, extensive research has been conducted on the validity and reliability of benefit transfer, including international benefit transfer (Dugstad & Navrud, 2023; Hynes et al., 2013; Lindhjem & Navrud, 2008; Ready & Navrud, 2006). Given the relatively close geographical and cultural proximity in this study, the likelihood of similar preferences across the islands is higher than in many other benefit transfer applications, especially those involving international transfers. Nonetheless, this assumption should be carefully evaluated in future research by assessing transfer errors. Still, this consideration should be carefully

examined in future research through examining transfer errors. Another consideration for future research is the temporal stability of preferences. The survey on residents of Maui was conducted in 2017. Preferences for recreational services could have temporal variability. This assumption has further implications for the predicted values of future coral loss under different SSP scenarios, given that preferences are assumed constant over time. Rolfe & Dyack, (2019) conducted a meta-analysis to test the temporal stability of recreation values, finding that TC values have increased up to 72 percent, and that the contingent valuation method is more reliable for benefit transfer over time than the TC method.

The ecological predictions are spatially explicit coral reef cover changes under different SSPs using the Atlantis ecological model. These projections are combined with economic valuations of changes in recreational services using a value function transfer approach. Modeling this welfare far into the future have inherent uncertainties with large ecosystem models using modeling such as Atlantis (Bracis et al., 2020). Disadvantages arise through the complex interactions of many parameters that true form can be unknown and is estimated or calibrated on the available data. The results, however, are in line with the literature exploring climate scenarios and coral reefs in the area (Lane et al., 2015; Matz et al., 2020). The Atlantis model reveal that some of the earliest and most significant declines in coral reef health are expected to occur in marine protected areas around the island of Hawai‘i. These areas, managed under the Marine Life Conservation District (MLCD) and Community-Based Subsistence Fishing Area (CBSFA) and are important for preserving biodiversity and supporting ecological resilience. These regions are vulnerable to environmental stressors, including climate change, overfishing, and pollution. The findings suggest additional adaptive strategies may be necessary to address emerging threats under even the most optimistic climate scenario. The importance to continuously evaluate and adapt existing management strategies will increase the resilience and long-term sustainability of Hawai‘i’s coral reef ecosystems. These results also provide insights that can inform the design of future marine protected areas and support broader regional and global conservation initiatives.

The analysis should be interpreted as an exploration of spatially dependent welfare changes to local populations, providing decision-makers with actionable insights into the social and economic trade-offs of coastal management under future climate scenarios. The estimates are intended to evaluate and identify welfare impacts across communities, with a specific focus on disadvantaged populations. This approach addresses policy-relevant environmental justice

concerns and highlights variation in welfare losses across different regions. The results reveal that disadvantaged communities, particularly on islands like Kaua‘i and Maui, experience higher welfare losses than non-disadvantaged communities. This disparity signals significant challenges, as these communities often have fewer resources to adapt or recover, underscoring the need to consider equity in loss and resilience. In other words, near-equity in welfare losses does not imply equal capacity to cope with or manage those losses. The findings emphasize the importance of prioritizing equity in resource management and policy development. Addressing these disparities requires targeted strategies to reduce disproportionate impacts on vulnerable communities. By tailoring management approaches to account for differences in vulnerability and adaptive capacity, policymakers can more effectively and equitably mitigate welfare losses in both disadvantaged and non-disadvantaged areas, aligning with broader environmental justice objectives.

Conclusion

By integrating welfare losses associated with recreational sites, this analysis highlights several key challenges. The welfare estimates account for both substitutable activities and site-specific attributes across recreational locations on the Hawaiian Islands, incorporating heterogeneity in ecological conditions and environmental characteristics at each site. These ecological attributes were then linked to projected changes in the nearshore environment under different climate scenarios to estimate welfare changes for Hawai‘i residents. The estimates also incorporate population growth projections. Under more optimistic scenarios (e.g., SSP1 and SSP2), both the United States and Hawai‘i are expected to experience population increases, which leads to higher aggregate welfare losses due to a greater number of individuals being affected by degraded environmental conditions. Moreover, as ecosystem services like coral reefs become scarcer relative to market goods and generational experience, the willingness to pay to protect or access them is likely to rise. These findings suggest even with emissions mitigation efforts substantial welfare losses is in store, particularly when accounting for ecological degradation, growing populations, and the increasing value of ecosystem services.

This paper integrates an ecosystem model that captures biophysical, economic, and social dimensions of the marine environment with a spatially explicit recreational valuation model applied to the Main Hawaiian Islands We demonstrate how declines in coral reef quality under various climate scenarios affect social welfare, accounting for the heterogeneous nature of

nearshore coastal recreation sites. A key strength of our approach lies in its spatial resolution. By simulating the localized impacts of climate change and management strategies across diverse geographic areas, the model provides decision-makers with actionable insights into the distributional welfare effects of alternative coastal policies. Furthermore, the analysis highlights the ethical dimensions of coastal management, emphasizing the need to consider equity and justice when evaluating trade-offs and outcomes.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research was supported by the National Oceanic and Atmospheric Administration's Ocean Acidification Program Regional Ocean Acidification Vulnerability and Resilience Assessment Grant Award # NA21OAR0170191.

Data availability

Data will be made available on request.

References

Andrello, M., Darling, E. S., Wenger, A., Suárez-Castro, A. F., Gelfand, S., & Ahmadia, G. N. (2022). A global map of human pressures on tropical coral reefs. *Conservation Letters*, *15*(1), e12858. https://doi.org/10.1111/conl.12858

Armstrong, C., Holen, S., Navrud, S., & Seifert, I. (2012). *The Economics of Ocean Acidification—A scoping study*. University of Tromsø.

Asner, G. P., Vaughn, N. R., Heckler, J., Knapp, D. E., Balzotti, C., Shafron, E., Martin, R. E., Neilson, B. J., & Gove, J. M. (2020). Large-scale mapping of live corals to guide reef conservation. *Proceedings of the National Academy of Sciences*, *117*(52), 33711–33718. https://doi.org/10.1073/pnas.2017628117

Audzijonyte, A., Pethybridge, H., Porobic, J., Gorton, R., Kaplan, I., & Fulton, E. A. (2019). Atlantis: A spatially explicit end-to-end marine ecosystem model with dynamically integrated physics, ecology and socio-economic modules. *Methods in Ecology and Evolution*, *10*(10), 1814–1819. https://doi.org/10.1111/2041-210X.13272

Bateman, I. J., Brouwer, R., Ferrini, S., Schaafsma, M., Barton, D. N., Dubgaard, A., Hasler, B., Hime, S., Liekens, I., Navrud, S., De Nocker, L., Ščeponavičiūtė, R., & Semėnienė, D. (2011). Making Benefit Transfers Work: Deriving and Testing Principles for Value Transfers for Similar and Dissimilar Sites Using a Case Study of the Non-Market Benefits of Water Quality Improvements Across Europe. *Environmental and Resource Economics*, *50*(3), 365–387. https://doi.org/10.1007/s10640-011-9476-8

Bateman, I. J., Harwood, A. R., Mace, G. M., Watson, R. T., Abson, D. J., Andrews, B., Binner, A., Crowe, A., Day, B. H., Dugdale, S., Fezzi, C., Foden, J., Hadley, D., Haines-Young, R., Hulme, M., Kontoleon, A., Lovett, A. A., Munday, P., Pascual, U., … Termansen, M. (2013). Bringing Ecosystem Services into Economic Decision-Making: Land Use in the United Kingdom. *Science*, *341*(6141), 45–50. https://doi.org/10.1126/science.1234379

Baumgärtner, S., Klein, A. M., Thiel, D., & Winkler, K. (2015). Ramsey Discounting of Ecosystem Services. *Environmental and Resource Economics*, *61*(2), 273–296. https://doi.org/10.1007/s10640-014-9792-x

Baustert, P., Othoniel, B., Rugani, B., & Leopold, U. (2018). Uncertainty analysis in integrated environmental models for ecosystem service assessments: Frameworks, challenges and gaps. *Ecosystem Services*, *33*, 110–123. https://doi.org/10.1016/j.ecoser.2018.08.007

Bracis, C., Lehuta, S., Savina-Rolland, M., Travers-Trolet, M., & Girardin, R. (2020). Improving confidence in complex ecosystem models: The sensitivity analysis of an Atlantis ecosystem model. *Ecological Modelling*, *431*, 109133. https://doi.org/10.1016/j.ecolmodel.2020.109133

Brander, L. M., Rehdanz, K., Tol, R. S. J., & Van Beukering, B., Pieter van. (2012). The economic impact of ocean acidification on coral reefs. *Climate Change Economics*, *3*(1), 1–29.

Brander, L. M., Van Beukering, P., & Cesar, H. S. J. (2007). The recreational value of coral reefs: A meta-analysis. *Ecological Economics*, *63*(1), 209–218. https://doi.org/10.1016/j.ecolecon.2006.11.002

Cesar, H. S. J. (Herman S. J.), & Beukering, P. V. (2004). Economic valuation of the coral reefs of Hawai'i. *Pacific Science*, *58*(2), 231–242. https://doi.org/10.1353/psc.2004.0014

Chen, P.-Y., Chen, C.-C., Chu, L., & McCarl, B. (2015). Evaluating the economic damage of climate change on global coral reefs. *Global Environmental Change*, *30*, 12–20. https://doi.org/10.1016/j.gloenvcha.2014.10.011

Colt, S. G., & Knapp, G. P. (2016). Economic Effects of an Ocean Acidification Catastrophe. *American Economic Review*, *106*(5), 615–619. https://doi.org/10.1257/aer.p20161105

Costa, B., & Kendall, M. S. (2016). *Marine biogeographic assessment of the main Hawaiian Islands: A collaborative investigation* (BOEM 2016-035; NOAA Technical Memorandum NOS NCCOS). NOAA. https://repository.library.noaa.gov/view/noaa/12906

Costanza, R., Kubiszewski, I., Stoeckl, N., & Kompas, T. (2021). Pluralistic discounting recognizing different capital contributions: An example estimating the net present value

of global ecosystem services. *Ecological Economics*, *183*, 106961. https://doi.org/10.1016/j.ecolecon.2021.106961

Díaz, S., Settele, J., Brondízio, E. S., Ngo, H. T., Agard, J., Arneth, A., Balvanera, P., Brauman, K. A., Butchart, S. H. M., Chan, K. M. A., Garibaldi, L. A., Ichii, K., Liu, J., Subramanian, S. M., Midgley, G. F., Miloslavich, P., Molnár, Z., Obura, D., Pfaff, A., … Zayas, C. N. (2019). Pervasive human-driven decline of life on Earth points to the need for transformative change. *Science*, *366*(6471), eaax3100. https://doi.org/10.1126/science.aax3100

Drupp, M. A., Hänsel, M. C., Fenichel, E. P., Freeman, M., Gollier, C., Groom, B., Heal, G. M., Howard, P. H., Millner, A., Moore, F. C., Nesje, F., Quaas, M. F., Smulders, S., Sterner, T., Traeger, C., & Venmans, F. (2024). Accounting for the increasing benefits from scarce ecosystems. *Science*, *383*(6687), 1062–1064. https://doi.org/10.1126/science.adk2086

Dugstad, A., Grimsrud, K. M., Kipperberg, G., Lindhjem, H., & Navrud, S. (2021). Scope Elasticity of Willingness to pay in Discrete Choice Experiments. *Environmental and Resource Economics*, *80*(1), 21–57. https://doi.org/10.1007/s10640-021-00577-7

Dugstad, A., & Navrud, S. (2023). The Reliability of Delphi Surveys and Benefit Transfer to Predict Outcomes of Contingent Valuation Surveys. *Land Economics*, *99*(3), 413–432. https://doi.org/10.3368/le.99.3.051922-0037R

Dundas, S. J., & Von Haefen, R. H. (2020). The Effects of Weather on Recreational Fishing Demand and Adaptation: Implications for a Changing Climate. *Journal of the Association of Environmental and Resource Economists*, *7*(2), 209–242. https://doi.org/10.1086/706343

EPA, U. S. E. P. A. (2024). *EJScreen Technical Documentation.*

Estrada, F., & Botzen, W. J. W. (2021). Economic impacts and risks of climate change under failure and success of the Paris Agreement. *Annals of the New York Academy of Sciences*, *1504*(1), 95–115. https://doi.org/10.1111/nyas.14652

Fabricius, K. E., Langdon, C., Uthicke, S., Humphrey, C., Noonan, S., De'ath, G., Okazaki, R., Muehllehner, N., Glas, M. S., & Lough, J. M. (2011). Losers and winners in coral reefs acclimatized to elevated carbon dioxide concentrations. *Nature Climate Change*, *1*(3), 165–169. https://doi.org/10.1038/nclimate1122

Fernandes, J. A., Papathanasopoulou, E., Hattam, C., Queirós, A. M., Cheung, W. W. W. L., Yool, A., Artioli, Y., Pope, E. C., Flynn, K. J., Merino, G., Calosi, P., Beaumont, N., Austen, M. C., Widdicombe, S., & Barange, M. (2017). Estimating the ecological, economic and social impacts of ocean acidification and warming on UK fisheries. *Fish and Fisheries*, *18*(3), 389–411. https://doi.org/10.1111/faf.12183

Ferrario, F., Beck, M. W., Storlazzi, C. D., Micheli, F., Shepard, C. C., & Airoldi, L. (2014). The effectiveness of coral reefs for coastal hazard risk reduction and adaptation. *Nature Communications*, *5*(1), 3794. https://doi.org/10.1038/ncomms4794

Ferrini, S., Schaafsma, M., & Bateman, I. J. (2015). Ecosystem Services Assessment and Benefit Transfer. In R. J. Johnston, J. Rolfe, R. S. Rosenberger, & R. Brouwer (Eds.), *Benefit Transfer of Environmental and Resource Values* (Vol. 14, pp. 275–305). Springer Netherlands. https://doi.org/10.1007/978-94-017-9930-0_13

Fezzi, C., Bateman, I. J., & Ferrini, S. (2014). Using revealed preferences to estimate the Value of Travel Time to recreation sites. *Journal of Environmental Economics and Management*, *67*(1), 58–70. https://doi.org/10.1016/j.jeem.2013.10.003

Fezzi, C., Ford, D. J., & Oleson, K. L. L. (2023). The economic value of coral reefs: Climate change impacts and spatial targeting of restoration measures. *Ecological Economics*, *203*, 107628. https://doi.org/10.1016/j.ecolecon.2022.107628

Frieler, K., Meinshausen, M., Golly, A., Mengel, M., Lebek, K., Donner, S. D., & Hoegh-Guldberg, O. (2013). Limiting global warming to 2 °C is unlikely to save most coral reefs. *Nature Climate Change*, *3*(2), 165–170. https://doi.org/10.1038/nclimate1674

Fu, X., Kakkar, D., Chen, J., Moynihan, K. M., Hegland, T. A., & Blossom, J. (2023). A COMPARATIVE STUDY OF METHODS FOR DRIVE TIME ESTIMATION ON GEOSPATIAL BIG DATA: A CASE STUDY IN USA. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *XLVIII-4/W7-2023*, 53–60. https://doi.org/10.5194/isprs-archives-XLVIII-4-W7-2023-53-2023

Fulton, E. A., Fuller, M., Smith, A. D. M., & Punt, A. E. (2004). Ecological indicators of the ecosystem effects of fishing. *Australian Fisheries Management Authority Report*, *R99*(1546), 116.

Giraud, T. (2022). osrm: Interface Between R and the OpenStreetMap-BasedRouting Service OSRM. *Journal of Open Source Software*, *7*(78), 4574. https://doi.org/10.21105/joss.04574

Good, A. M., & Bahr, K. D. (2021). The coral conservation crisis: Interacting local and global stressors reduce reef resiliency and create challenges for conservation solutions. *SN Applied Sciences*, *3*(3), 312. https://doi.org/10.1007/s42452-021-04319-8

Haab, T. C., & McConnell, K. E. (2002). *Valuing Environmental and Natural Resources: The Econometrics of Non-market Valuation*. Edward Elgar Publishing. https://doi.org/10.4337/9781843765431

Hafezi, M., Sahin, O., Stewart, R. A., Connolly, R. M., Mackey, B., & Ware, D. (2020). Adaptation strategies for coral reef ecosystems in Small Island Developing States: Integrated modelling of local pressures and long-term climate changes. *Journal of Cleaner Production*, *253*, 119864. https://doi.org/10.1016/j.jclepro.2019.119864

Hafezi, M., Stewart, R. A., Sahin, O., Giffin, A. L., & Mackey, B. (2021). Evaluating coral reef ecosystem services outcomes from climate change adaptation strategies using integrative system dynamics. *Journal of Environmental Management*, *285*, 112082. https://doi.org/10.1016/j.jenvman.2021.112082

Hansen, B. E. (2022). *Econometrics*. Princeton University Press.

Harris, D. L., Rovere, A., Casella, E., Power, H., Canavesio, R., Collin, A., Pomeroy, A., Webster, J. M., & Parravicini, V. (2018). Coral reef structural complexity provides important coastal protection from waves under rising sea levels. *Science Advances*, *4*(2), eaao4350. https://doi.org/10.1126/sciadv.aao4350

Hoegh-Guldberg, O., Mumby, P. J., Hooten, A. J., Steneck, R. S., Greenfield, P., Gomez, E., Harvell, C. D., Sale, P. F., Edwards, A. J., Caldeira, K., Knowlton, N., Eakin, C. M., Iglesias-Prieto, R., Muthiga, N., Bradbury, R. H., Dubi, A., & Hatziolos, M. E. (2007). Coral Reefs Under Rapid Climate Change and Ocean Acidification. *Science*, *318*(5857), 1737–1742. https://doi.org/10.1126/science.1152509

Hoegh-Guldberg, O., Poloczanska, E. S., Skirving, W., & Dove, S. (2017). Coral Reef Ecosystems under Climate Change and Ocean Acidification. *Frontiers in Marine Science*, *4*. https://www.frontiersin.org/articles/10.3389/fmars.2017.00158

Hynes, S., Norton, D., & Hanley, N. (2013). Adjusting for Cultural Differences in International Benefit Transfer. *Environmental and Resource Economics*, *56*(4), 499–519. https://doi.org/10.1007/s10640-012-9572-4

IUCN, (Int. Union Conserv. Nature). (2018). *The IUCN Red List of Threatened Species* [Dataset]. Gland, Switz.

Jennings, V., San Antonio, K. M., Brown, M. J., Choice, L., Simpson, Q., Ford, I., Cho, H. J., Solis, P., Lacey, A., & Robinson, D. (2024). Place-Based Conservation in Coastal and Marine Ecosystems: The Importance of Engagement with Underrepresented Communities. *Sustainability*, *16*(22), 9965. https://doi.org/10.3390/su16229965

Jiang, L., O'Neill, B. C., Zoraghein, H., & Dahlke, S. (2020). Population scenarios for U.S. states consistent with shared socioeconomic pathways. *Environmental Research Letters*, *15*(9), 094097. https://doi.org/10.1088/1748-9326/aba5b1

Johnston, R. J., & Wainger, L. A. (2015). Benefit Transfer for Ecosystem Service Valuation: An Introduction to Theory and Methods. In R. J. Johnston, J. Rolfe, R. S. Rosenberger, & R. Brouwer (Eds.), *Benefit Transfer of Environmental and Resource Values* (Vol. 14, pp. 237–273). Springer Netherlands. https://doi.org/10.1007/978-94-017-9930-0_12

Kleypas, J., & Yates, K. (2009). Coral Reefs and Ocean Acidification. *Oceanography*, *22*(4), 108–117. https://doi.org/10.5670/oceanog.2009.101

Lane, D., Jones, R., Mills, D., Wobus, C., Ready, R. C., Buddemeier, R. W., English, E., Martinich, J., Shouse, K., & Hosterman, H. (2015). Climate change impacts on freshwater fish, coral reefs, and related ecosystem services in the United States. *Climatic Change*, *131*(1), 143–157. https://doi.org/10.1007/s10584-014-1107-2

Leggett, C. G., Scherer, N., Haab, T. C., Bailey, R., Landrum, J. P., & Domanski, A. (2018). Assessing the Economic Benefits of Reductions in Marine Debris at Southern California Beaches: A Random Utility Travel Cost Model. *Marine Resource Economics*, *33*(2), 133–153. https://doi.org/10.1086/697152

Lindhjem, H., & Navrud, S. (2008). How reliable are meta-analyses for international benefit transfers? *Ecological Economics*, *66*(2–3), 425–435. https://doi.org/10.1016/j.ecolecon.2007.10.005

Lopes, A. F., & Whitehead, J. C. (2023). Estimating the Ex Ante Recreational Loss of an Oil Spill Using Revealed and Stated Preference Data. *Land Economics*, *99*(4), 490–508. https://doi.org/10.3368/le.99.4.102022-0084R

Matz, M. V., Treml, E. A., & Haller, B. C. (2020). Estimating the potential for coral adaptation to global warming across the Indo-West Pacific. *Global Change Biology*, *26*(6), 3473–3481. https://doi.org/10.1111/gcb.15060

McClanahan, T. R. (2024). Reconsidering and rescaling climate change predictions for coral reefs. *Nature Climate Change*, *14*(8), 779–781. https://doi.org/10.1038/s41558-024-02079-y

McLean, M., Cuetos-Bueno, J., Nedlic, O., Luckymiss, M., & Houk, P. (2016). Local Stressors, Resilience, and Shifting Baselines on Coral Reefs. *PLOS ONE*, *11*(11), e0166319. https://doi.org/10.1371/journal.pone.0166319

Moore, C., & Fuller, J. (2022). Economic Impacts of Ocean Acidification: A Meta-Analysis. *Marine Resource Economics*, *37*(2), 201–219. https://doi.org/10.1086/718986

Mullen, H., Whyte, K., & Holifield, R. (2023). Indigenous Peoples and the Justice40 Screening Tool: Lessons from EJSCREEN. *Environmental Justice*, *16*(5), 360–369. https://doi.org/10.1089/env.2022.0045

Munday, P. L., Jones, G. P., Pratchett, M. S., & Williams, A. J. (2008). Climate change and the future for coral reef fishes. *Fish and Fisheries*, *9*(3), 261–285. https://doi.org/10.1111/j.1467-2979.2008.00281.x

Narita, D., Rehdanz, K., & Tol, R. S. J. (2012). Economic costs of ocean acidification: A look into the impacts on global shellfish production. *Climatic Change*, *113*(3–4), 1049–1063. https://doi.org/10.1007/s10584-011-0383-3

Newell, R. G., & Pizer, W. A. (2003). Discounting the distant future: How much do uncertain rates increase valuations? *Journal of Environmental Economics and Management*, *46*(1), 52–71. https://doi.org/10.1016/S0095-0696(02)00031-1

Nordhaus, W. D. (2010). Economic aspects of global warming in a post-Copenhagen environment. *Proceedings of the National Academy of Sciences*, *107*(26), 11721–11726. https://doi.org/10.1073/pnas.1005985107

Nordhaus, W. D. (2017). Revisiting the social cost of carbon. *Proceedings of the National Academy of Sciences*, *114*(7), 1518–1523. https://doi.org/10.1073/pnas.1609244114

Office of Management & Budget, O. (2023, November 9). *Circular No. A-4*. https://www.whitehouse.gov/wp-content/uploads/2023/11/CircularA-4.pdf

Oleson, K. L. L., Bagstad, K. J., Fezzi, C., Barnes, M. D., Donovan, M. K., Falinski, K. A., Gorospe, K. D., Htun, H., Lecky, J., Villa, F., & Wong, T. M. (2020). Linking Land and Sea Through an Ecological-Economic Model of Coral Reef Recreation. *Ecological Economics*, *177*, 106788. https://doi.org/10.1016/j.ecolecon.2020.106788

O'Neill, B. C., Kriegler, E., Riahi, K., Ebi, K. L., Hallegatte, S., Carter, T. R., Mathur, R., & Van Vuuren, D. P. (2014). A new scenario framework for climate change research: The concept of shared socioeconomic pathways. *Climatic Change*, *122*(3), 387–400. https://doi.org/10.1007/s10584-013-0905-2

Pandolfi, J. M., Connolly, S. R., Marshall, D. J., & Cohen, A. L. (2011). Projecting Coral Reef Futures Under Global Warming and Ocean Acidification. *Science*, *333*(6041), 418–422. https://doi.org/10.1126/science.1204794

Pascal, N., Allenbach, M., Brathwaite, A., Burke, L., Le Port, G., & Clua, E. (2016). Economic valuation of coral reef ecosystem service of coastal protection: A pragmatic approach. *Ecosystem Services*, *21*, 72–80. https://doi.org/10.1016/j.ecoser.2016.07.005

Pathak, A., Van Beynen, P. E., Akiwumi, F. A., & Lindeman, K. C. (2021). Impacts of climate change on the tourism sector of a Small Island Developing State: A case study for the Bahamas. *Environmental Development*, *37*, 100556. https://doi.org/10.1016/j.envdev.2020.100556

Peng, M., & Oleson, K. L. L. (2017). Beach Recreationalists' Willingness to Pay and Economic Implications of Coastal Water Quality Problems in Hawaii. *Ecological Economics*, *136*, 41–52. https://doi.org/10.1016/j.ecolecon.2017.02.003

Phaneuf, D. J., & Smith, V. K. (2005). Chapter 15 Recreation Demand Models. In *Handbook of Environmental Economics* (Vol. 2, pp. 671–761). Elsevier. https://doi.org/10.1016/S1574-0099(05)02015-2

Ready, R., & Navrud, S. (2006). International benefit transfer: Methods and validity tests. *Ecological Economics*, *60*(2), 429–434. https://doi.org/10.1016/j.ecolecon.2006.05.008

Reguero, B. G., Storlazzi, C. D., Gibbs, A. E., Shope, J. B., Cole, A. D., Cumming, K. A., & Beck, M. W. (2021). The value of US coral reefs for flood risk reduction. *Nature Sustainability*, *4*(8), 688–698. https://doi.org/10.1038/s41893-021-00706-6

Rolfe, J., & Dyack, B. (2019). Testing Temporal Stability of Recreation Values. *Ecological Economics*, *159*, 75–83. https://doi.org/10.1016/j.ecolecon.2019.01.016

Rosenberger, R. S., & Loomis, J. B. (2017). Benefit Transfer. In P. A. Champ, K. J. Boyle, & T. C. Brown (Eds.), *A Primer on Nonmarket Valuation* (Vol. 13, pp. 431–462). Springer Netherlands. https://doi.org/10.1007/978-94-007-7104-8_11

Souter, D., Planes, S., Wicquart, J., Obura, D., & Staub, F. (2021). *Status of Coral Reefs of the World: 2020*. International Coral Reef Initiative. https://doi.org/10.59387/WOTJ9184

Spalding, M., Burke, L., Wood, S. A., Ashpole, J., Hutchison, J., & Zu Ermgassen, P. (2017). Mapping the global value and distribution of coral reef tourism. *Marine Policy*, *82*, 104–113. https://doi.org/10.1016/j.marpol.2017.05.014

Speers, A. E., Besedin, E. Y., Palardy, J. E., & Moore, C. (2016). Impacts of climate change and ocean acidification on coral reef fisheries: An integrated ecological–economic model. *Ecological Economics*, *128*, 33–43. https://doi.org/10.1016/j.ecolecon.2016.04.012

Stern, N. (2007). *The Economics of Climate Change: The Stern Review* (1st ed.). Cambridge University Press. https://doi.org/10.1017/CBO9780511817434

Suggett, D. J., Edwards, M., Cotton, D., Hein, M., & Camp, E. F. (2023). An integrative framework for sustainable coral reef restoration. *One Earth*, *6*(6), 666–681. https://doi.org/10.1016/j.oneear.2023.05.007

van Hooidonk, R., Maynard, J., Tamelander, J., Gove, J., Ahmadia, G., Raymundo, L., Williams, G., Heron, S. F., & Planes, S. (2016). Local-scale projections of coral reef futures and implications of the Paris Agreement. *Scientific Reports*, *6*(1), 39666. https://doi.org/10.1038/srep39666

Wedding, L. M., Pittman, S. J., Lepczyk, C. A., Parrain, C., Puniwai, N., Boyle, J. S., Goldberg, E. G., Young, M., Marty, P., Wilhelm, K., Taylor, S., & Crowder, L. B. (2024). Integrating the multiple perspectives of people and nature in place-based marine spatial planning. *Npj Ocean Sustainability*, *3*(1), 43. https://doi.org/10.1038/s44183-024-00071-9

Weijerman, M. (2020). *Development of an Atlantis mode for Hawaii to support ecosystem-based management*. https://doi.org/10.25923/CWQB-1Z04

Whitehead, J. C., Phaneuf, D. J., Dumas, C. F., Herstine, J., Hill, J., & Buerger, B. (2010). Convergent Validity of Revealed and Stated Recreation Behavior with Quality Change: A Comparison of Multiple and Single Site Demands. *Environmental and Resource Economics*, *45*(1), 91–112. https://doi.org/10.1007/s10640-009-9307-3

Zoraghein, H., & O'Neill, B. C. (2020). U.S. State-level Projections of the Spatial Distribution of Population Consistent with Shared Socioeconomic Pathways. *Sustainability*, *12*(8), 3374. https://doi.org/10.3390/su12083374

Figures

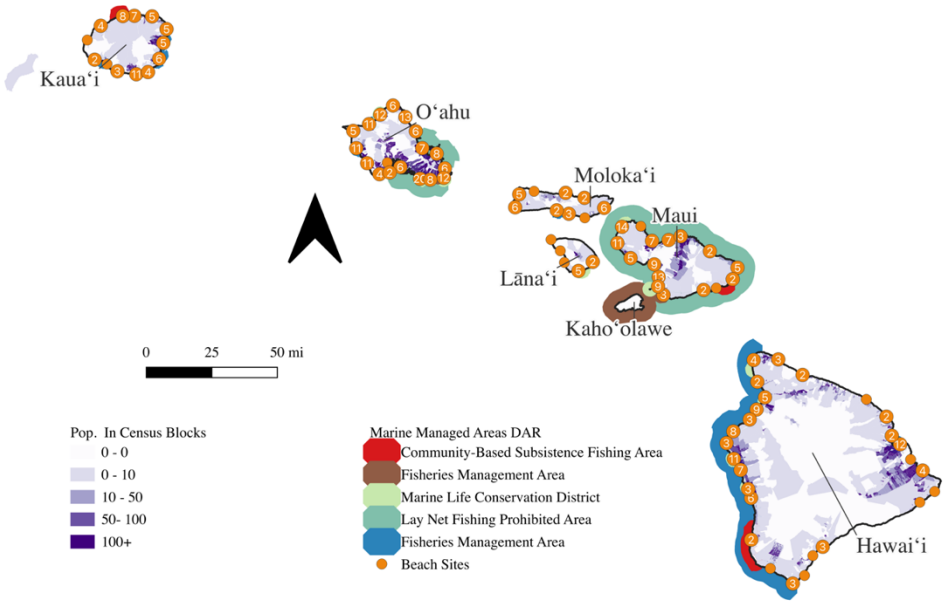

Figure 1. Coastal Sites across the main Hawaiian Islands. (N = 444) *Note;* Clustering of coastal sites are represented by numbers in red dots to better represent coastal and reduce overlapping. Population represented in 2020 census blocks. Designated Marine Managed Areas (MMAs) across the Main Hawaiian Islands, as identified by the Hawai'i Division of Aquatic Resources (DAR). The boundaries reflect varying levels of management and permitted activities, highlighting the spatial overlap between conservation efforts and recreational coastal use.

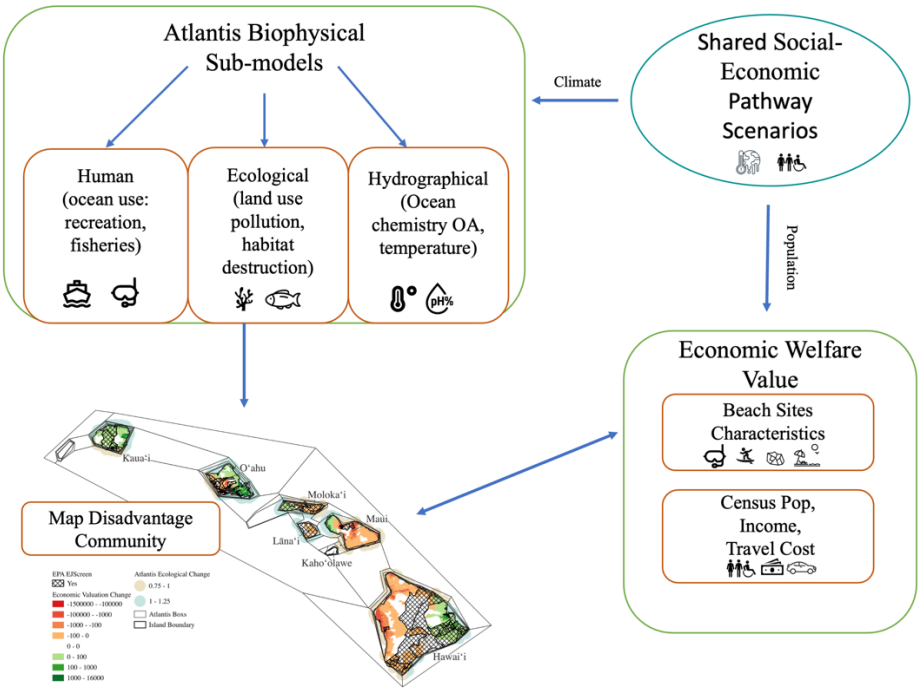

Figure 2. Integrated Modeling Conceptual Diagram: Atlantis predicts ecological changes incorporating sub-models of human, ecological, and hydrographical changes estimated spatially for each poylgons. Nearshore changes are matched to coastal site locations. Welfare estimates are the compensating variation from predicted changes from Atlantis within the nearshore environment.

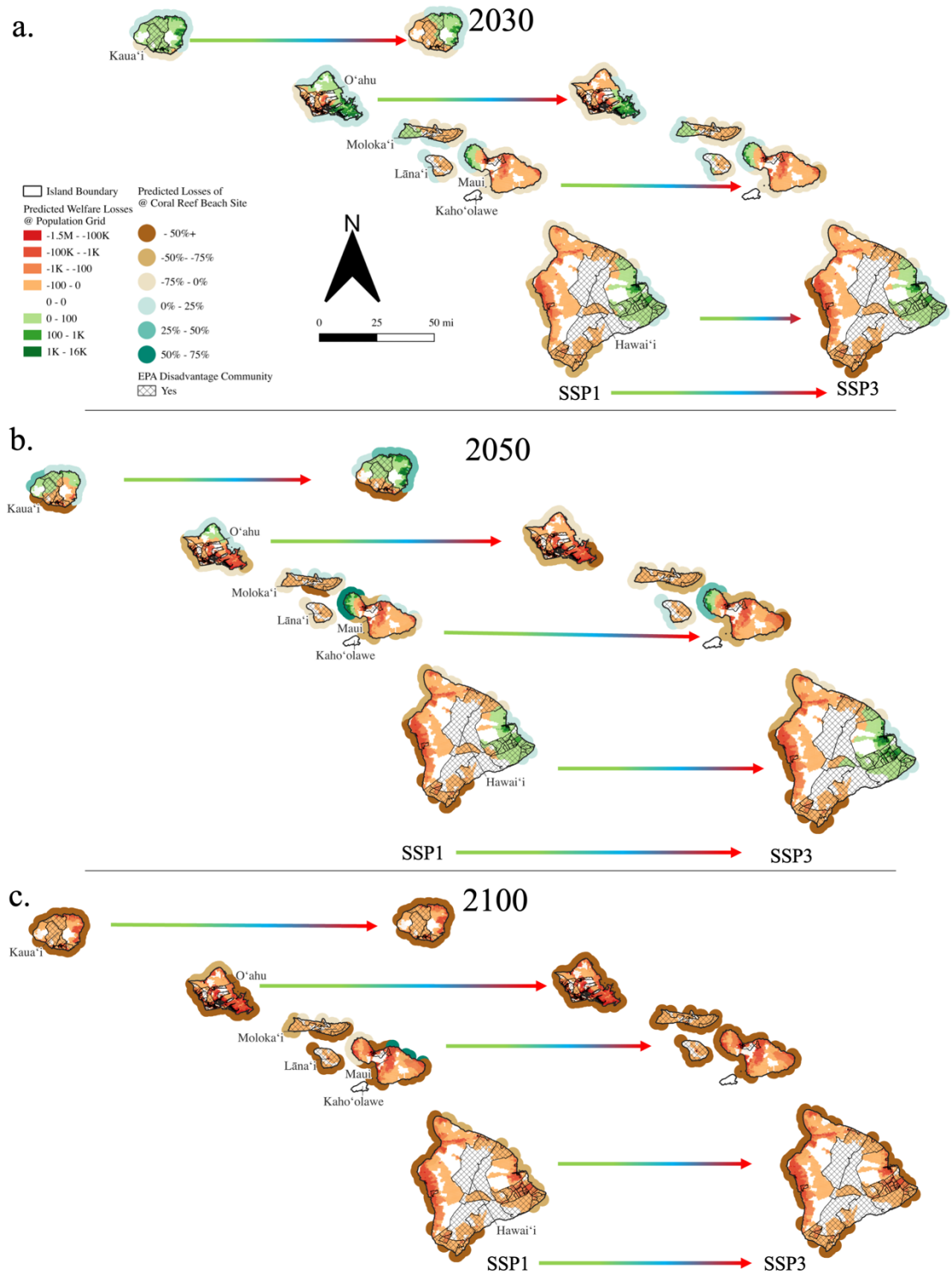

Figure 3 represents the MHI in (a) SSP1 (left) and (b) SSP3(right) in snap shots which correspond to timeperiods from top to bottom starting at 2030, 2050 (mid-century) and 2100 (end-century)Projected welfare change is presented on land, while coral cover change is indicated offshore. Disadvataged communities are identified in grided overlay.

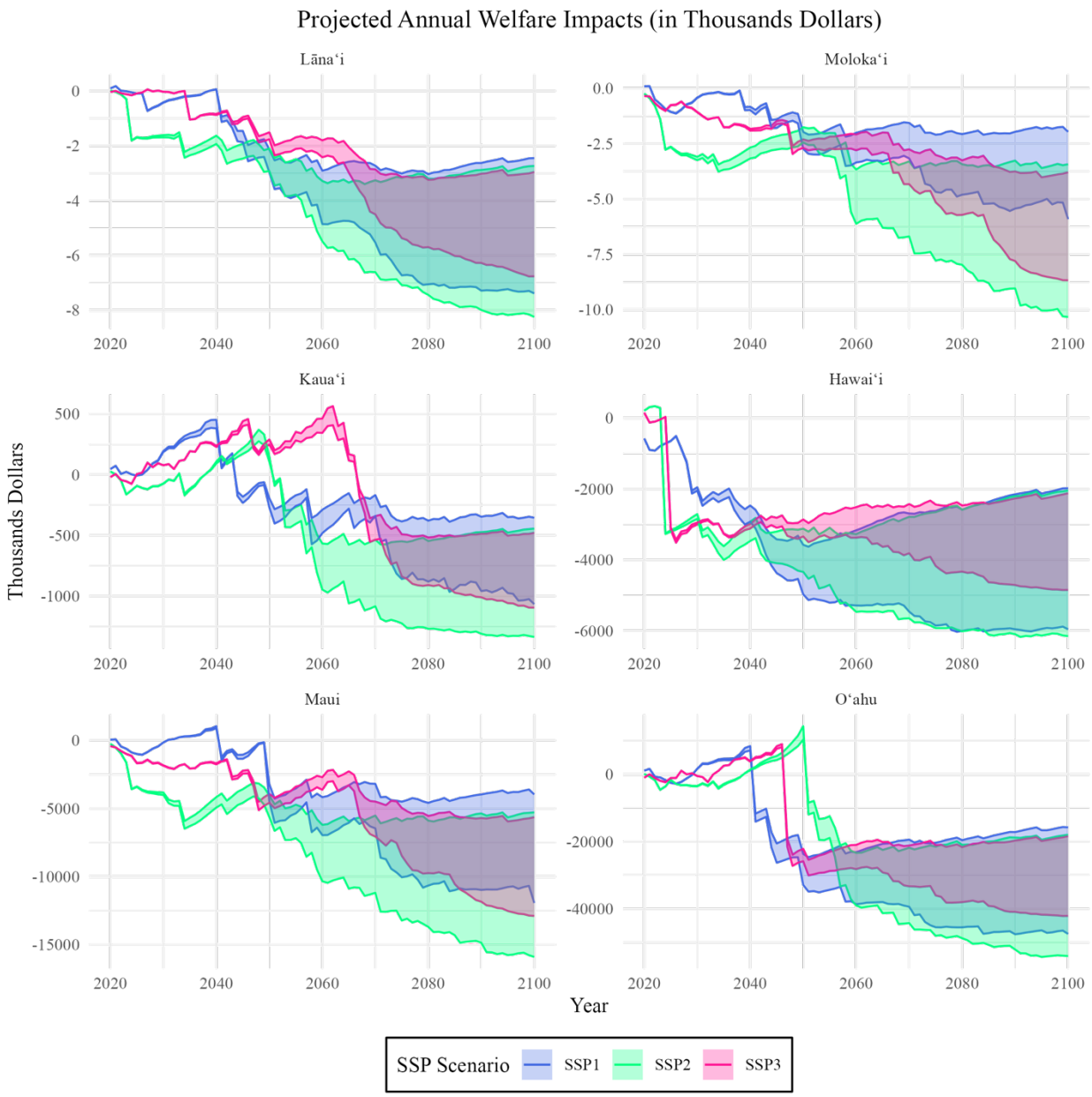

Figure 4 Projected Annual Welfare impacts from projected changes in coral reefs under future climate scenarios for Hawaii Residents across each island to the year 2100. Upper bound represents a declining discount rate; lower bound represents a pluralistic discounting. Green line represents the welfare estimates under the climate scenario SSP1, blue under climate scenario SSP2, the red line represents the welfare estimates under climate scenario SSP3.

Tables

Island	Maui (Original Study)	O‘ahu	Kaua‘i	Hawai‘i (Big Island)	Lana‘i	Moloka‘i
Distant per island (Grid * Sites)	191,948	257,174	95,542	1,089,184	4675	22533
Avg. distance (Mi) to Sites	34.9	28.1	32.5	63.6	14.5	20.6
Medium Income	86,872	87,722	82,818	65,401	59,053	47,329
Pop. Total 18 Older	122K	810K	51K	159K	2.5K	5.5K
EJ Population %	8.5%	11.2%	2%	31%	100%	100%
Coral Reef Cover %	11.5	4.64	4.75	12.9	8.61	6.14
Avg. Resource Fish Biomass g/m2	16.4	5.62	18.7	15.8	4.38	12.2
Coastal Site in Meters	404	1801	2071	518	1073	781
Num. Coastal Recreation	94	149	62	101	10	28
Num. Snorkel	60	64	26	46	2	8
Num. Surf	39	91	24	23	2	3

Table 1: Descriptive statistics of islands Maui, O‘ahu, Kaua‘i, Hawai‘i, Lana‘i and Moloka‘i. Distance per island is the number of distance pairings between population grid cells and identified coastal recreation sites. Avg. distance to sites represents the mean distance in miles from population centroids to coastal recreation sites. Socioeconomic data include median income and adult population size (18+), while the percentage of the population identified as disadvantaged is based on EJScreen indicators. Ecological variables include average coral reef cover and resource fish biomass (g/m²). Recreational site metrics include the total coastal length (in meters) and counts of coastal recreation, snorkeling, and surfing sites verified across spatial datasets. Maui serves as the reference island for value function estimation.

Table 2.A Disadvantaged Communities Average Welfare Loss Per Person in 2024$ Annual

	SSP1			SSP3		
Island	**2030**	**2050**	**2100**	**2030**	**2050**	**2100**
Hawai‘i	-0.97	-3.39	-5.04	-1.57	-1.97	-5.99
Kaua‘i	1.92	-4.81	-17.00	0.21	1.73	-20.30
Lāna‘i	-0.09	-0.68	-1.79	-0.01	-0.49	-2.15
Maui	-21.22	-80.38	-155.56	-28.10	-89.72	-207.14
Moloka‘i	-0.06	-0.61	-1.42	-0.15	-0.67	-2.55
O‘ahu	-1.91	-48.64	-83.52	-4.69	-45.80	-94.62
Average	**-3.72**	**-23.09**	**-44.06**	**-5.72**	**-22.82**	**-55.46**

B. All Other Communities Average Predicted Welfare Loss Per Person

	SSP1			SSP3		
Island	**2030**	**2050**	**2100**	**2030**	**2050**	**2100**
Hawai‘i	-16.89	-42.09	-52.59	-24.24	-46.94	-68.52
Kaua‘i	3.43	-3.73	-20.95	1.40	9.19	-35.12
Lāna‘i						
Maui	-3.57	-31.25	-89.43	-15.86	-58.82	-170.18

Moloka‘i						
O‘ahu	4.18	-56.08	-90.19	1.07	-68.35	-140.61
Average	**-5.68**	**-33.29**	**-63.29**	**-9.41**	**-41.23**	**-103.61**

Table 2 Snapshot of Predicted Average Annual Welfare Losses By Identified Disadvantaged Community per KM square grid. Reported average values per KM square in 2024$ by identified as historical and currently disadvantaged under the classification of the EJ screen. Highlighted values in A. note the on average higher impacts within islands respective disadvantaged communities than other communities.